

Making AI Work for All Americans

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Laboratory for Information and Decision Systems (LIDS)
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By: Chara Podimata, Luca Carlone, Marzyeh Ghassemi,
Priya L. Donti, Sertac Karaman, and Swati Gupta

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Executive Summary

Artificial intelligence (AI) has advanced very rapidly in recent years. These advances are grounded in machine learning (ML), where extremely-large models learned from unstructured data have displayed unprecedented human-like ability on a variety of tasks. At this point, it may seem that leadership in AI basically boils down to training the largest ML models. Instead, we argue that establishing American leadership will require focusing on further **foundational and algorithmic advances** that close the gap in key qualities like **robustness, resilience, and explainability** and **transition these advances directly into applications**, in order to **ensure AI can develop and transform many sectors of the American economy**.

Sustaining and enhancing America’s leadership in AI fundamentally relies on **ensuring that AI technologies benefit all Americans**, fitting the needs and requirements of a multitude of applications and economic sectors. For instance, AI technologies developed for use in areas such as electric power systems and autonomous transportation must be inherently safe and robust, in order to avoid safety-critical failures such as large-scale power outages or vehicle crashes [M+2019, DK2021]. AI developed for financial and healthcare applications must be inherently privacy-preserving, in order to avoid costly and damaging leaks to personal consumer data [K+2023]. And AI developed for legal and governmental applications must be inherently interpretable and explainable, to ensure the alignment of decision-making processes with legal rules and requirements [DK2017, R2019].

While certain classes of AI technologies, such as large language models (LLMs), have found substantial commercial success, these technologies do not satisfy the above requirements, and thereby only fit the needs of a very narrow set of applications relative to AI’s significantly wider potential. Importantly, requirements such as safety, robustness, privacy preservation, interpretability, and explainability are not simply “add-ons” to existing AI technologies.

Developing technologies that meet these requirements demands a fundamentally ground-up and heterogeneous approach to AI innovation, supported by (a) large-scale investments in basic and applied research, (b) enabling infrastructure such as improved data, simulation infrastructure, compute access, and cross-sectoral innovation hubs, and (c) agile identification of needs and ongoing evaluation of the state of AI technologies relative to these needs, in order to continuously shape necessary oversight and incentives. Such investments to date have been critical in enabling the development and ultimate commercial success of AI technologies such as LLMs, and similar continued investments are likewise critical for replicating this success across a wider range of AI paradigms and reaping the associated societal and commercial benefits.

In service of the above goals, this response provides recommendations in three key areas:

1. **Supporting basic research and innovation across heterogeneous AI areas:** Large-scale investments in basic and applied research, across a heterogeneous range of AI areas, will be critical to developing the next generation of technologies satisfying on-the-ground requirements such as safety, robustness, privacy, interpretability, and explainability.

2. **Improving enablers such as data, simulation, compute, & collaboration:** Sustaining and scaling American AI innovation and leadership requires improvements to the underlying innovation ecosystem, spanning data access and sharing, simulation infrastructure for scientific and engineering applications, and AI compute infrastructure, in a way that enables participation from a large and broad range of stakeholders.
3. **Establishing an independent AI regulatory body to foster innovation and oversight:** We live in a world where human actions across the economy are subject to oversight to benefit and protect the American people, for example, in healthcare, finance, and transportation sectors. At the very least, AI must not be excluded from the same oversight to which we subject humans. The regulatory framework for AI can be complex, and balancing innovation and the protection of the American people will be intricate. We recommend focusing AI regulation towards the agile identification of needs and requirements for AI technologies, as well as assessment and oversight of AI technologies in alignment with these requirements, to provide ongoing input to AI innovation strategies and ensure the success of developed technologies.

I. Enabling AI to Work for All Americans via a Heterogeneous Approach

AI has the potential to transform a wide range of societal and commercial applications, across sectors such as heterogeneous as electric power systems, agriculture, transportation, and medicine. However, this potential has yet to be realized for the vast majority of applications, as **today’s AI does not meet many of the fundamental requirements of these applications**. For instance, safety and robustness are essential requirements when using AI in safety-critical infrastructure such as electric power systems and in many robotic systems [M+2019, DK2021, G+2024]; privacy preservation is critical in applications involving individual consumer data (such as in healthcare and finance) [K+2023]; and interpretability and explainability are crucial for governmental and legal applications that require public accountability [DK2017, R2019].

Developing AI technologies that meet these requirements will require large-scale research and innovation across a wide range of AI and machine learning (ML) paradigms, including, for example, physics-informed ML,¹ safe ML,² privacy-preserving ML,³ interpretable and explainable ML,⁴ neuro-symbolic AI,⁵ probabilistic and Bayesian ML,⁶ and tiny ML.⁷ Importantly, these paradigms are not simply “retrofits” to existing AI techniques – instead, **AI techniques must be designed from the ground-up with these requirements built in**. This is analogous to the notion of privacy-by-design in software development, where intentional up-front software design choices are necessary to ensure that sensitive data will remain protected, whereas retroactive attempts to make non-secure software secure rarely succeed [I2024]. Unlike in privacy-by-design software development, however, there are as-of-yet no established “playbooks” to making AI private by design. Similarly, approaches to make AI safe, interpretable, explainable, probabilistic, physics-informed, and/or energy efficient *by design* are **active albeit early-stage areas of research, requiring substantial and concerted investment in basic and applied research to advance AI technologies in these directions**. Such investment is likely to yield significant downstream societal and commercial benefits by yielding AI technologies that are well-poised to catalyze progress across a broad range of societal sectors.

¹ *Physics-informed ML* is a branch of machine learning that focuses on integrating physics-based equations and/or physics-based simulations with or within ML techniques.

² *Safe ML* refers to the development and deployment of ML models that prioritize robustness, security, and reliability to guarantee trustworthy AI decision-making.

³ *Privacy-preserving ML* aims to ensure that de-identified data used to train ML models cannot be re-identified or reverse engineered, via techniques such as (e.g.) differential privacy.

⁴ *Interpretable ML* refers to ML techniques whose internals can be easily understood or audited by algorithm developers, users, or regulators. *Explainable ML* refers to ML techniques that are able to provide (potentially post-hoc) justifications for their outputs.

⁵ *Neuro-symbolic AI* combines data-driven machine learning techniques with knowledge-guided (symbolic) techniques, in order to obtain the joint benefits of pattern recognition and logical reasoning approaches.

⁶ *Probabilistic and Bayesian ML* refer to techniques that are able to capture and express some notion of uncertainty and/or confidence in their outputs.

⁷ *Tiny ML* is a branch of machine learning that focuses on deploying models on ultra-low-power, resource-constrained devices, such as microcontrollers and embedded systems.

Importantly, just as private-by-design software can look fundamentally different from software that is not private-by-design, AI techniques look fundamentally different depending on the requirements with which they are designed. For instance, designing safe ML approaches may entail embedding safety criteria within neural network architectures [D2022], designing risk-sensitive ML training techniques [GF2015, B+2021], or combining techniques from ML with techniques from control theory [G+2024] or formal verification [FP2018]. As another example, designing interpretable ML approaches may entail intentionally constructing machine learning models whose weights are easy to analyze, rather than (e.g.) black-box neural network architectures [R+2022]. Accordingly, **America’s approach to AI development must be fundamentally adaptive and application-driven, rather than reliant on a singular class of models such as foundation models or large language models (LLMs).** We do not expect AI benefits to trickle down from a select set of methods; instead, AI must be developed bottom-up with specific application needs in mind [R+2024]. Additionally, AI development must engage a broad coalition of stakeholders, including academia, civil society, and public institutions, to foster foundational methodological advances and an approach that fundamentally centers the needs of all Americans.

In the rest of this section, we provide a few concrete, selected examples of important directions for AI innovation. These are not meant to be exhaustive, but rather, indicative of the heterogeneous approach to AI that is necessary to serve the needs of different societal and commercial applications. We conclude with selected recommendations on how to support the development of heterogeneous AI approaches.

Example: AI in Robotics and Autonomous Vehicles. Robotics and autonomous systems are positioned to revolutionize and transform a wide range of commercial and societal applications. For instance, robots are widely used in retail and logistics, where companies such as Amazon have already deployed >750K units, and where warehouse automation was a \$7B market in 2024 that is expected to grow to a \$54B market by 2030 [Z2024]. Agricultural robotics is well-positioned to support increased and more efficient food production, with autonomous systems being used for functions such as weeding, seeding, harvesting, spraying, and milking; precision agriculture is expected to become \$17B market by 2030 (~\$6B in 2023) [F2025]. Finally, autonomous transportation carries the promise of saving lives and largely reducing the social and economic costs of transportation; for instance, in 2022, >42K people died in US car crashes (>1.2M worldwide), with autonomous vehicles providing the potential to reduce such crashes by 90% [C2024].

Despite this incredible potential, robotics and autonomous systems require fundamental advances to ensure a full and positive impact. The “Cambrian explosion” we are observing in robotics is the result of better hardware (e.g., motors and robotic hands), better algorithms (importantly, including AI and ML), more data (due to more deployed robots and self-driving cars), and more compute resources. At the same time, robotics poses fundamentally different challenges for AI than other “standard” applications. For instance, widespread applications of AI, including

LLM-based applications like ChatGPT, often fail to produce reasonable answers when faced with out-of-distribution inputs. While this is often acceptable when chatting with an AI agent, it can put human lives at risk when it comes to robotics applications. Fundamental research is necessary to tackle safe and trustworthy AI of salience to robotics, including topics such as AI assurance and robustness, fault detection and isolation, resilience, communication, and control.

Example: AI in Electric Power Systems. Electricity is a critical backbone of modern American society, but is facing challenges with respect to affordability and reliability, due to factors such as aging infrastructure and extreme weather events. AI technologies have the potential to transform the operation of electric power systems in ways that significantly enhance affordability and reliability, by providing faster and more scalable algorithms for the scheduling and control of power generation, batteries, and flexible demand, thus enabling the power grid to be operated more efficiently and in a way that is increasingly adaptive to unexpected events [DK2021].

Despite this potential, AI technologies today face several limitations that preclude their deployment in safety-critical aspects of power grid operations and control. For instance, AI outputs are not guaranteed to be physically feasible (e.g., respect the laws of physics regarding how power flows on a power grid), which has the potential to lead to large-scale power outages, economic losses, and loss of lives. Progress in physics-informed and safe ML, including safe reinforcement learning, is thus critical to enabling the use of AI for the optimization and control of electric power systems [S+2024]. AI in power systems also fundamentally operates on hardware-based systems, rather than purely in software, requiring developments in tiny ML, hardware integration of ML, and ML techniques that can handle factors such as sensor noise, communication latency, and actuation latency [SG2023, S+2023]. Transmission congestion in power systems is also becoming an increasingly large problem, costing billions of dollars per year [W2024]; to address this, AI is being considered for use in applications such as topology optimization, which requires reasoning over billions of topology-switching actions. New neuro-symbolic AI techniques [GL2023] for power systems may be critical to enabling this large-scale reasoning, by allowing intelligent search over the large space of topology actions.

Basic and applied research in physics-informed ML, tiny ML, hardware-integrated ML, and neuro-symbolic AI is therefore critical to enabling the use of AI in power systems. In addition to research funding, the development of grid simulation infrastructure and hardware-integrated testbeds is critical to enabling the development and testing of AI approaches in power grids.

Example: AI in Healthcare. Learning from health data requires models that are robust to changes in time (new populations), place (new hospitals), and manner (new treatments), rather than fit to points (humans) who lie in the center of the distribution. Models must also work for humans in the tails of health distributions, which requires general improvements to machine learning, both in improving the efficiency of methods, and addressing the downstream gaps created by non-robust models. In health, demographic attributes like age, gender and race have historically been used to improve the average performance of clinical models and scores, but their inclusion has led to over- or under-estimation of risk. Given the demonstrated gaps that occur with the

naive inclusion of demographic attributes in state of the art models in health, there must be strong criteria for inclusion of such attributes in health – for instance, by explicitly including per-demographic performance gap constraints [SGU2023]. Importantly “state of the art” methods like differential privacy and distributionally robust optimization have been shown to perform poorly in health settings specifically, as the methods do not scale well to data with heavy tails or attribute shifts [S+2021, Y+2023]. There must therefore be ongoing efforts to pinpoint actionable barriers to model performance, such as model uncertainty due to data complexity and quality.

Example: Navigating Mixed-Fidelity Data. AI is increasingly being used to generate fake information, combine data sources with low-quality and high-quality signals, and mimic human interactions with decision systems. These AI outputs are then often used as inputs to other AI models, such as those in critical decision-making systems ranging across supply chains, transportation, and finance. In other words, the quality of data on which AI approaches are trained can be extremely unreliable. This raises significant challenges in terms of navigating fake data, discounting fake sources of information when training models, and learning from behaviors that look “human” but are machine-generated. In addition to opening a huge opportunity and market for high-quality data, this also demands fundamental research towards the development of AI systems that can detect fake and/or low-quality signals.

To foster the development of AI technologies that can meet the needs of heterogeneous societal and commercial applications, we provide several key recommendations.

Recommendations:

- **Invest in basic and applied research to accelerate the development of heterogeneous AI approaches**, including (but not limited to) physics-informed ML, safe ML, privacy-preserving ML, interpretable and explainable ML, neuro-symbolic AI, probabilistic and Bayesian ML, and tiny ML. This should include funding for both broad-based application-agnostic AI research *and* specific application-driven AI research, both of which are critical to ensuring that AI methods are able to meet real-world challenges.
- **Fund interdisciplinary and cross-sectoral collaborations** where domain experts (in, e.g., power systems and healthcare) work alongside AI researchers to align AI development with societal needs. These funding mechanisms should allow financial support for both research institutions and deployment partners (e.g., small businesses or civil society organizations) who are collaborators on the work.
- **Fund application-specific enabling infrastructure, such as simulation environments and test beds for scientific and engineering applications**, that enable the development of heterogeneous AI approaches. For example, developing AI applications for the power sector can be significantly accelerated and improved via the availability of software simulation environments of the power grid as well as hardware-integrated testbeds.

II. Improving Data, Simulation, Compute, and Other Enablers

Data access is a critical enabler of AI research and development, and restrictive data practices disproportionately benefit large corporations while limiting academic and civil society research. Without open, transparent data-sharing mechanisms, academic institutions lose the ability to audit, validate, and improve AI models; this imbalance could significantly hinder national AI competitiveness. Likewise, broad-based access to computational infrastructure is fundamental to enabling a wide range of stakeholders to participate in the development of AI technologies, which is critical both to enhancing the volume of overall AI development and to advancing a heterogeneous set of AI technologies that serve the needs of a wide range of applications and sectors (see Section I).

Recommendation: Improve Data Access and Governance

- Provide clear federal guidance to streamline compliance with privacy and security regulations, enabling secure but accessible data-sharing frameworks.
- Expand federal data-sharing initiatives through agencies such as CMS and VA, ensuring that public-sector data is more accessible to researchers.
- Increase enforcement of data-sharing requirements (e.g., mandate open-access publication of federally funded health research datasets).
- Reduce barriers to AI R&D on federally controlled data by expanding cloud access and funding agreements that cover compute resources and compliance costs.
- Establish public-private data partnerships to create new AI-relevant datasets from sources like search engines and social media platforms, ensuring they are accessible for public-interest AI research.
- Expand programs like NIH's AIM-AHEAD to fund private-sector data curation initiatives and develop independent AI evaluation centers for healthcare applications.
- Fund training programs for de-identification experts to reduce bottlenecks in AI-driven health research while maintaining strong privacy protections.
- Establish sector-specific data task forces (e.g., in energy, manufacturing, and healthcare) to identify data gaps, access barriers, and necessary incentives for data sharing.

Recommendation: Improve Access to Computation and Simulation Infrastructure

- Provide affordable, scalable cloud computing resources to academic researchers, civil society, and small-to-medium enterprises to enable broad-based AI research and development.
- Fund the creation and maintenance of simulation tools and AI testbeds for developing AI solutions in scientific, engineering, and safety-critical sectors (e.g., for power systems, transportation, and industrial applications).

III. Fostering Innovation & Oversight via an Independent Regulatory Body

Human activity is regulated across major sectors of the economy, such as healthcare, finance, transportation, agriculture, manufacturing, and more, in order to ensure American people are protected. AI activity in such sectors will inevitably be regulated. It would be counterproductive to regulate human activity, but not hold our AI tools up to similar standards. Hence, whether or not AI should be regulated is not the right question, since some regulation of AI is inevitable. The question is: What should be the focus and the limits of AI regulation? The goal is to ensure America leads in AI innovation as well as the protection of the American people from all aspects of AI, whether they are developed domestically or internationally.

We believe that establishing American leadership in AI requires mechanisms to identify societal needs and requirements in an *agile manner*, in order to continuously shape necessary oversight and incentives, and to evaluate the alignment of specific AI systems with necessary requirements. To this end, we propose the creation of an independent AI oversight body, the Artificial Intelligence Regulatory Commission (AIRC), modeled on the Food and Drug Administration (FDA) but adapted for AI regulation. This agency would ensure that AI systems meet safety, reliability, and accountability standards. Proactive regulatory oversight in turn has the capability to spur further innovation—as has been the case for the FDA [FDA2024] – by enabling the ongoing identification of gaps in AI capabilities and subsequent prioritization of these areas for research and innovation funding. Establishing an independent AI regulatory body is critical to achieving this balance between safety and innovation, by providing structured oversight, evaluating AI technologies against evolving safety and accountability standards, and shaping ongoing innovation strategies

Our recommendation is to **create an Artificial Intelligence Regulatory Commission (AIRC)**, with key responsibilities including **AI auditing and maintaining a best-practices repository**, **AI licensing and risk-based regulation**, and **lifecycle-based AI evaluation**. In the remainder of the section, we expand upon each of these responsibilities.

The first and most important responsibility of such a regulatory body would be the **establishment and maintenance of standardized AI auditing practices**. As AI systems increasingly influence high-stakes domains, including healthcare, autonomous transportation, and financial decision-making, a centralized repository of best practices for AI auditing and evaluation would provide transparency and accountability. Ensuring that audit methodologies are **rigorous, publicly accessible**, and **consistently updated** would not only mitigate risks but also encourage AI developers to integrate safety and fairness measures from the outset ([CRB22], [C24], [LL+24]).

Regulation must also adopt a risk-based approach, ensuring that oversight is proportionate to the potential societal impact of different AI systems. A **licensing framework for AI systems**,

similar to FDA approvals for medical devices, could ensure that only AI technologies meeting robust safety and ethical standards are deployed in critical sectors. Similar proposals have been adopted for both dataset [GMV+18] and machine learning models [MW+2019]. In addition, leaders from the tech sector should be active participants in shaping the licensing framework; industry involvement is critical in ensuring adherence to required reporting mechanisms and maintaining transparency in AI development.

To further support AI development while maintaining regulatory oversight, the creation of **regulatory sandboxes** [OECD2023] would provide controlled environments where AI systems can be tested under real-world conditions before widespread deployment. These sandboxes would allow developers to refine AI models and address potential safety concerns while enabling regulators to assess risks and establish sector-specific guidelines. Additionally, independent oversight mechanisms must be integrated into the licensing process, ensuring that governments, consumer advocacy groups, and watchdog organizations can request audits of AI systems when concerns arise. By enabling independent audits, this framework would reinforce transparency and accountability, ensuring that AI technologies align with public interest while avoiding regulatory capture or undue industry influence

Continuous oversight is essential to maintaining accountability as AI systems evolve. Unlike static regulatory approaches that may quickly become outdated, AI regulation must be dynamic, adapting to technological advancements and emerging risks. By implementing **lifecycle-based evaluations**, regulators can ensure that AI systems remain aligned with safety, privacy, and fairness requirements long after their initial deployment. Drawing from established best practices in other regulated sectors, such as pharmaceutical and aviation safety, this approach would provide a structured yet flexible mechanism for AI governance.

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